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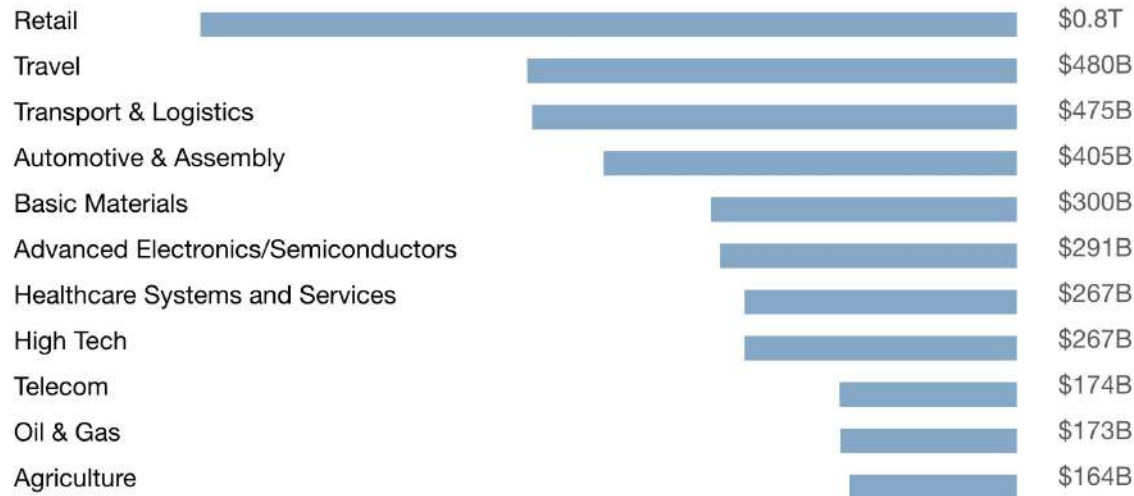
What is AI

Introduction

Introduction

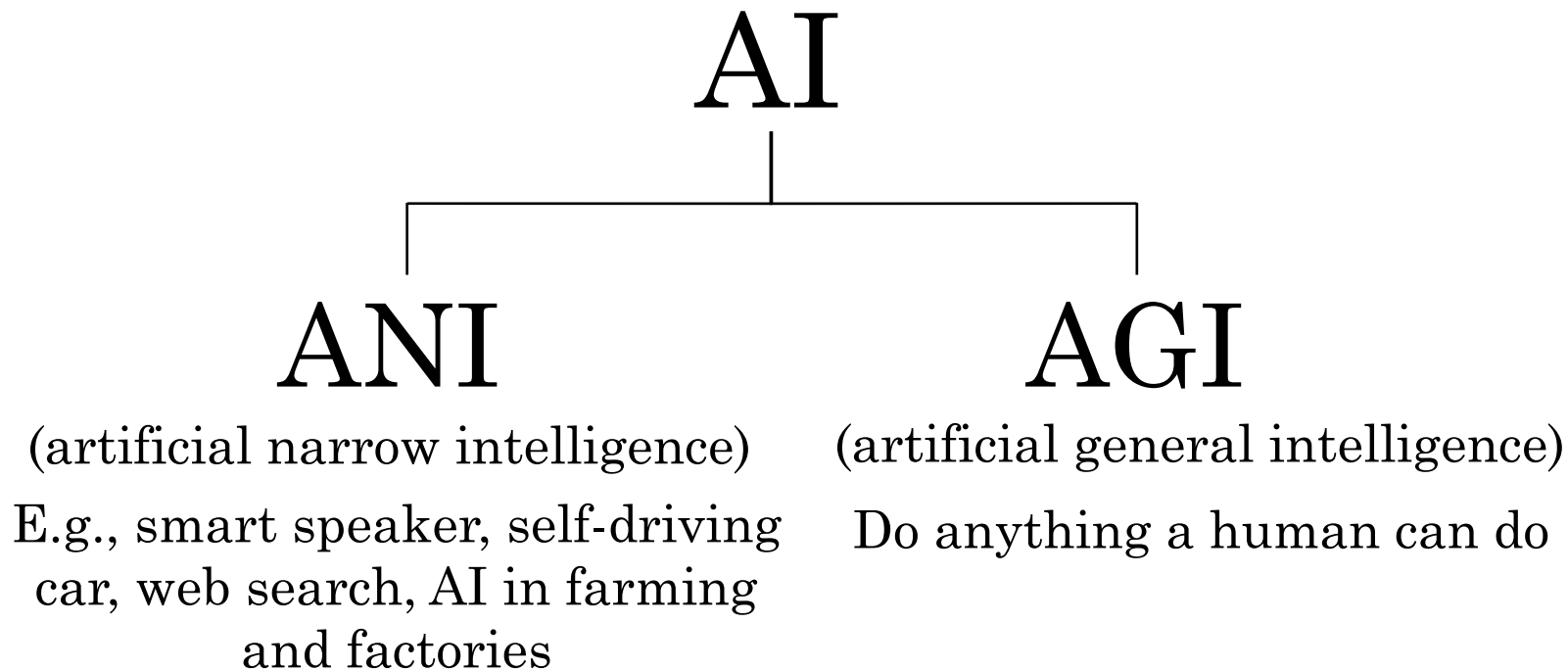
AI value creation
by 2030

**\$13
trillion**



[Source: McKinsey Global Institute.]

Demystifying AI



What you'll learn

- What is AI?
 - Machine Learning
 - Data
 - What makes an AI company
 - What machine learning can and cannot do
 - Optional: Intuitive explanation of Deep Learning
- Building AI projects
- Building AI in your company
- AI and society



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What is AI

Machine Learning

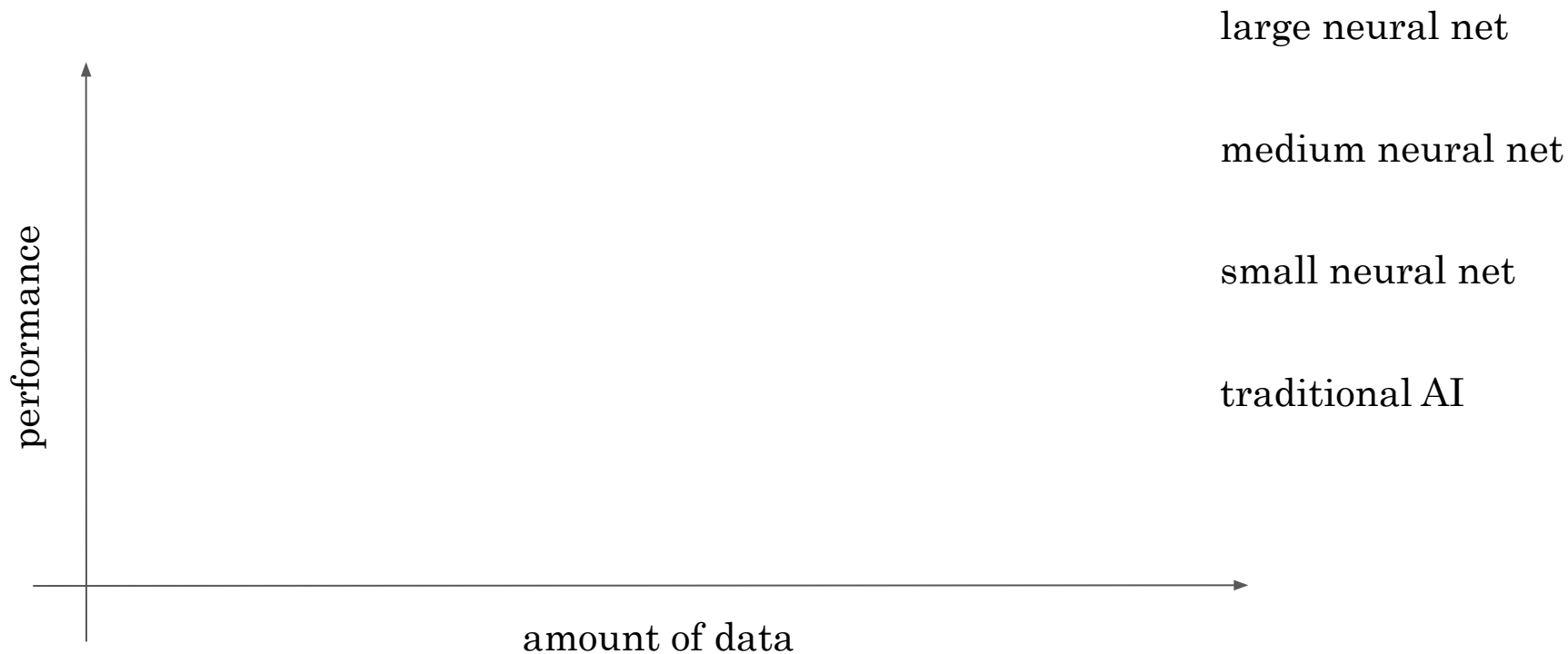
Supervised Learning

Input

Output

Input (A)	Output (B)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Chinese	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	Self-driving car
image of phone	defect? (0/1)	visual inspection

Why Now?





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What is AI

What is data

Example of a table of data (dataset)

house (square feet)	# of bedrooms	price (1000\$)
523	1	100
645	1	150
708	2	200
1034	3	300
2290	4	350
2545	4	440

image	label
	cat
	not cat
	cat
	not cat

“Google cat”

Acquiring data

- Manual labeling



cat



not
cat



cat



not
cat

- From observing user behaviors

user ID	time	price (\$)	purchased
4783	Jan 21 08:15.20	7.95	yes
3893	March 3 11:30.15	10.00	yes
8384	June 11 14:15.05	9.50	no
0931	Aug 2 20:30.55	12.90	yes

machine	temperature (°C)	pressure (psi)	machine fault
17987	60	7.65	N
34672	100	25.50	N
08542	140	75.50	Y
98536	165	125.00	Y

- Download from websites / partnerships

Use and mis-use of data

Don't throw data
at an AI team and
assume it will be
valuable.



Data is messy

- Garbage in, garbage out
- Data problems
 - Incorrect labels
 - Missing values
- Multiple types of data
 - images, audio, text

house (square feet)	# of bedrooms	price (1000\$)
523	1	100
645	1	0.001
708	unknown	200
1034	3	unknown
unknown	4	350
2545	unknown	440





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What is AI

The terminology of AI

Machine learning vs. data science

Home
prices

size (square feet)	# of bedrooms	# of bathrooms	newly renovated	price (1000\$)
523	1	2	N	100
645	1	3	N	150
708	2	1	N	200
1034	3	3	Y	300
2290	4	4	N	350
2545	4	5	Y	440

Homes with 3 bedrooms are more expensive than homes with 2 bedrooms of a similar size.

Newly renovated homes have a 15% premium.

Running AI system
(e.g., websites / mobile app)

Machine learning vs. data science

Machine learning

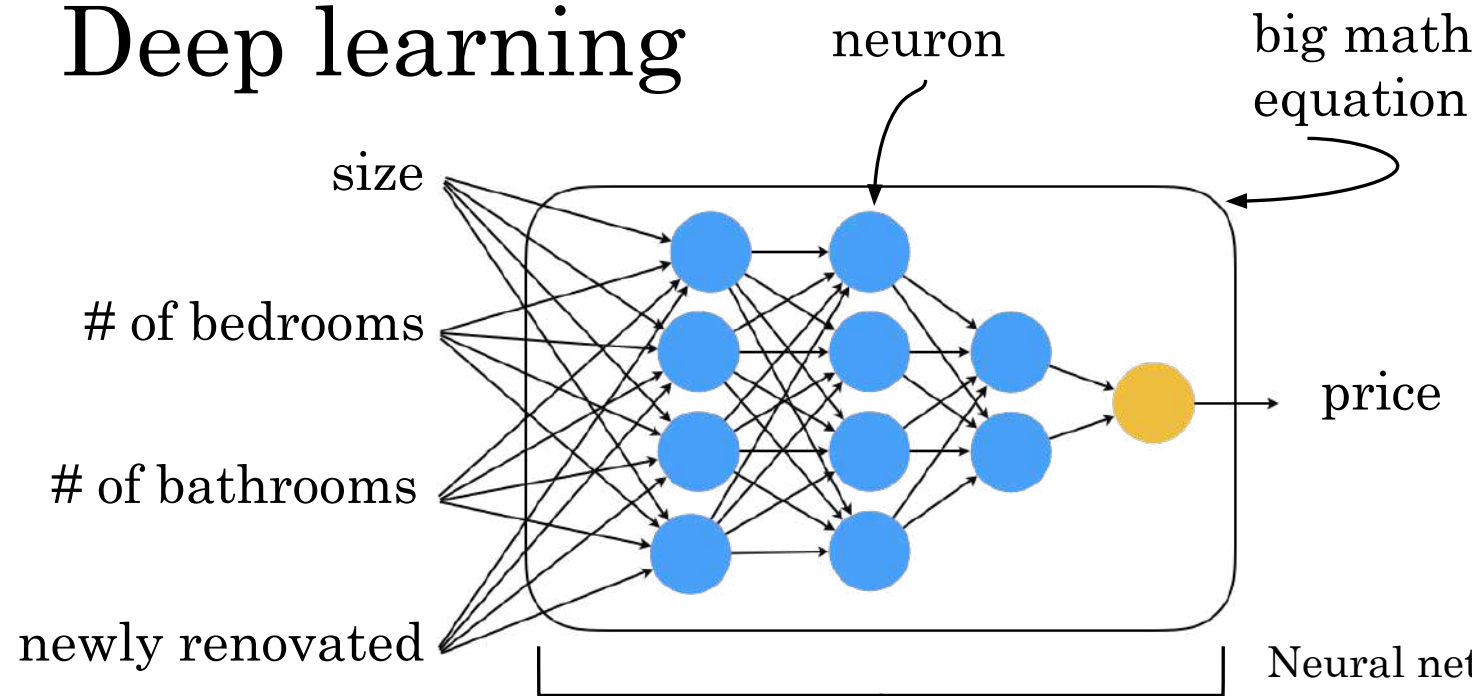
“Field of study that gives computers the ability to learn without being explicitly programmed.”

-Arthur Samuel (1959)

Data science

Science of extracting knowledge and insights from data.

Deep learning



(Artificial) Neural Network

Neural networks were originally inspired by the brain, but the details of how they work are almost completely unrelated to how biological brains work.

AI has many tools

- Machine learning and data science
- Deep learning / neural network
- Other buzzwords: Unsupervised learning, graphical models, planning, knowledge graph, ...



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What is AI

What makes an AI company?

A lesson from the rise of the Internet

Internet Era

Shopping mall + website
≠ Internet company

- A/B testing
- Short iteration time
- Decision making pushed down to engineers and other specialized roles

AI era

Any company + deep learning
≠ AI company

- Strategic data acquisition
- Unified data warehouse
- Pervasive automation
- New roles (e.g., MLE) and division of labor

AI Transformation

1. Execute pilot projects to gain momentum
2. Build an in-house AI team
3. Provide broad AI training
4. Develop an AI strategy
5. Develop internal and external communication



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What is AI

What machine learning
can and cannot do

Supervised Learning

Input (A)	Output (B)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Chinese	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	Self-driving car
image of phone	defect? (0/1)	visual inspection

Anything you can do with 1 second of thought,
we can probably now or soon automate.

What machine learning today can and cannot do

The toy arrived two days late, so I wasn't able to give it to my niece for her birthday.

Can I return it?



“Refund request”



Oh, sorry to hear that.
I hope your niece had a good birthday.
Yes, we can help with....

Input text

Refund/ Support/ Shipping

What happens if you try?

Input (A)

User email



Output (B)

2-3 paragraph response

1000 examples

“My box was damaged.”



Thank you for your email.

“Where do I write a review?”



Thank you for your email.

“What’s the return policy?”



Thank you for your email.

“When is my box arriving?”



Thank yes now your....

What makes an ML problem easier

1. Learning a “simple” concept
2. Lots of data available



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What is AI

More examples of what machine learning can and cannot do

Self-driving car

Can do



Cannot do



stop



hitchhiker



bike turn
left signal

1. Data
2. Need high accuracy

X-ray diagnosis



Can do

Diagnose pneumonia from
~10,000 labeled images

Cannot do

Diagnose pneumonia from
10 images of a medical textbook
chapter explaining pneumonia

Strengths and weaknesses of machine learning

ML tends to work well when:

1. Learning a “simple” function
2. There is lots of data available

ML tends to work poorly when:

1. Learning complex functions from small amounts of data
2. It is asked to perform on new types of data that it learned from





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What is AI

Non-technical explanation of
deep learning I (optional)

Demand prediction



price

demand

“neuron”

Demand prediction

price

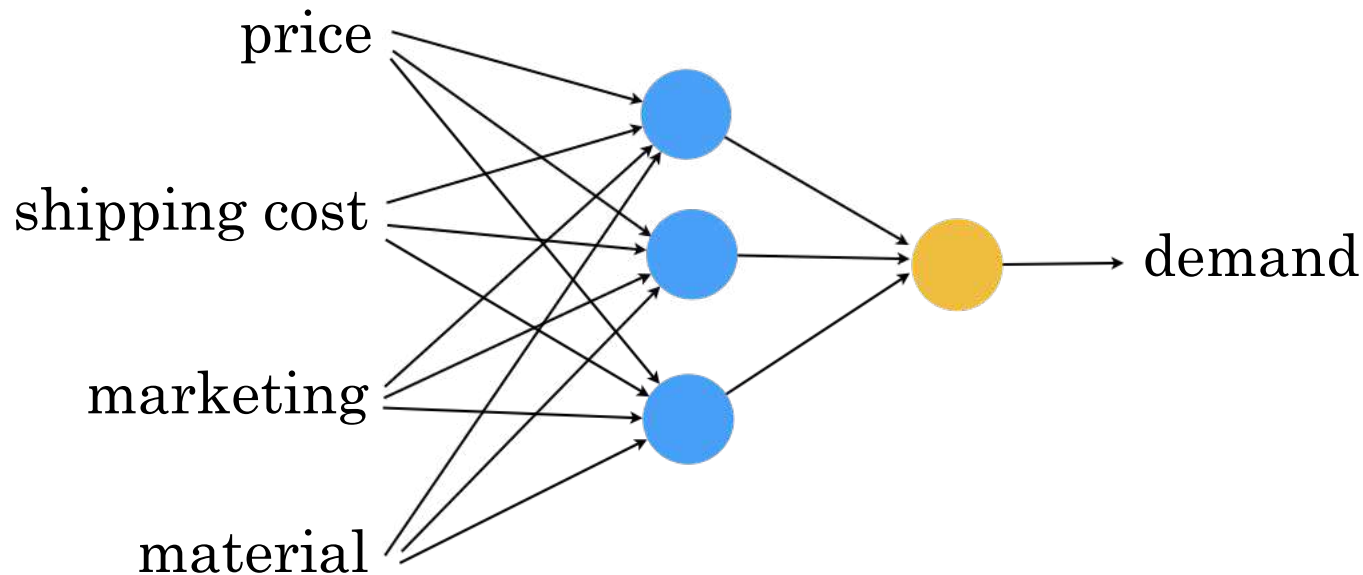
shipping cost

marketing

material

demand

Demand prediction





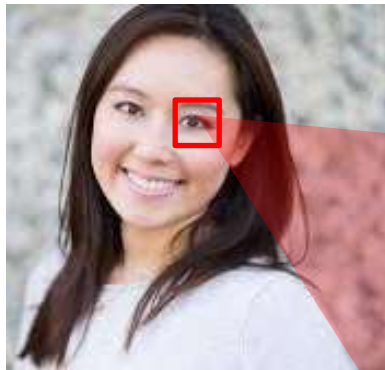
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What is AI

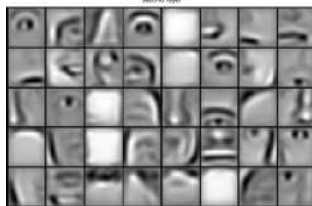
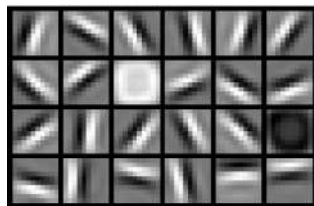
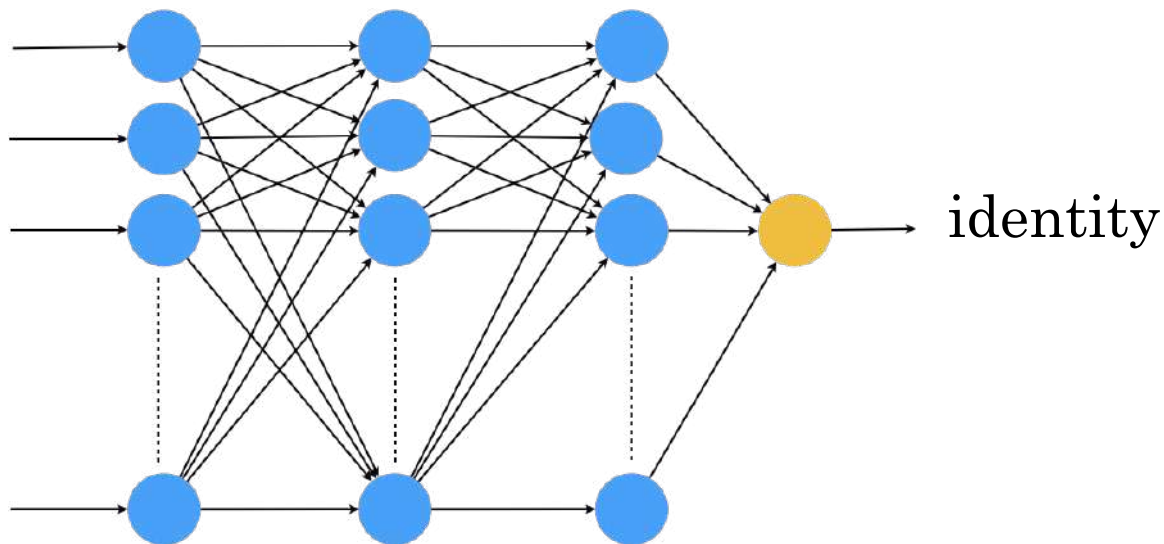
Non-technical explanation of
deep learning II (optional)

Face recognition



30	32	22	12	10	10	12	33	35	30
12	11	12	234	170	176	13	15	12	12
234	222	220	230	200	222	230	234	56	78
190	220	186	112	110	110	112	180	30	32
49	250	250	250	4	2	254	200	44	6
55	250	250	250	3	1	250	245	25	3
189	195	199	150	110	110	182	190	199	55
200	202	218	222	203	200	200	208	215	222
219	215	220	220	222	214	215	210	220	220
220	220	220	220	221	220	221	220	220	222

Face recognition



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Building AI Projects

Starting an AI project

Starting an AI project

- Workflow of projects
- Selecting AI projects
- Organizing data and team for the projects



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Building AI Projects

Workflow of a
machine learning project

Example: Speech recognition



Amazon
Echo / Alexa



Google
Home



Apple
Siri



Baidu
DuerOS

Key steps of a machine learning project

Echo / Alexa

1. Collect data

2. Train model

Iterate many times until
good enough

3. Deploy model

Get data back

Maintain / update model

Key steps of a machine learning project

Self-driving car

1. Collect data



image

position of other cars

2. Train model

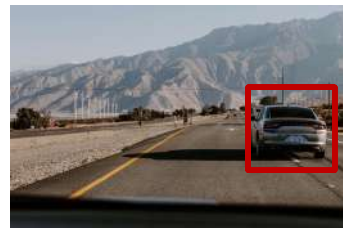
Iterate many times until
good enough



3. Deploy model

Get data back

Maintain / update model





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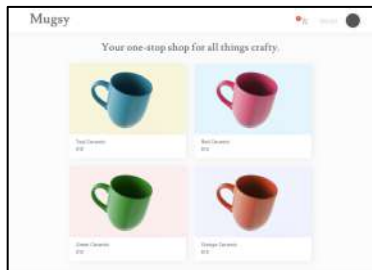
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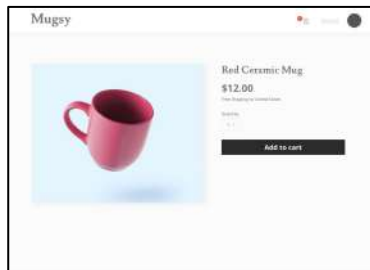
Workflow of a
data science project

Example: Optimizing a sales funnel

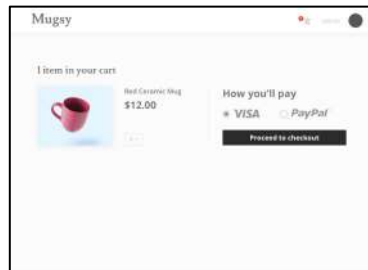
Visit website



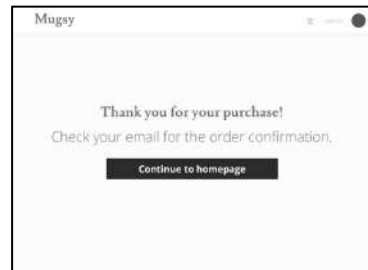
Product page



Shopping cart



Checkout



Key steps of a data science project

Optimizing a sales funnel

1. Collect data

User ID	Country	Time	Webpage
2009	Spain	08:34:30 Jan 5	home.html
2897	USA	13:20:22 May 18	redmug.html
4893	Philippines	22:45:16 Jun 11	mug.html

2. Analyze data

Iterate many times to get good insights

3. Suggest hypotheses/actions

Deploy changes

Re-analyze new data periodically

Key steps of a data science project

Manufacturing line

Mix clay



Shape mug



Add glaze



Fire kiln



Final inspection



Clay Batch #	Supplier	Mixing time (minutes)
001	ClayCo	35
034	GooClay	22
109	BrownStuff	28

1. Collect data

2. Analyze data

Iterate many times to get good insight

3. Suggest hypotheses/actions

Deploy changes

Re-analyze new data periodically

Mug Batch #	Country	Humidity	Temperature in kiln (F)	Duration in kiln (hours)
301	Spain	0.002%	1410°	22
302	USA	0.003%	1520°	24
303	Malaysia	0.002%	1420°	22



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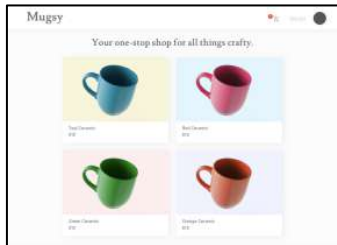
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Every job function
needs to learn how to use data

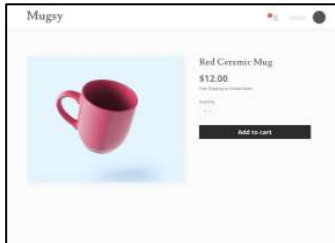
Sales

Data science

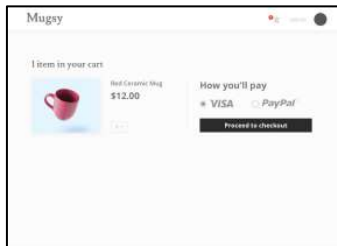
Visit website



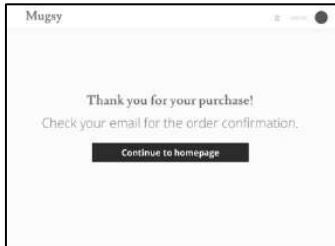
Product page



Shopping cart



Checkout



Optimize sales funnel

Machine learning

Name	Title	Company size	Email	Priority
Tayler	CEO	3050	tay@a..	high
Janet	Manager	230	jan@b..	medium
David	Intern	30	dave@c..	low

Automated lead sorting

Manufacturing line manager

Data science

Mix clay



Shape mug



Add glaze



Fire kiln



Final inspection



Optimize sales funnel

Machine learning



ok



ok

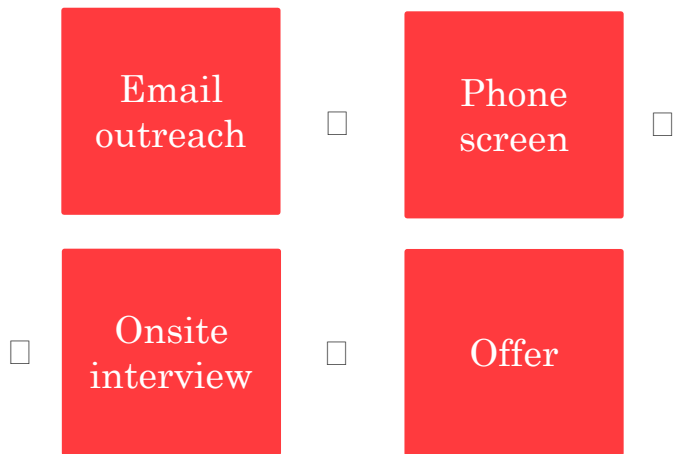


defect

Automated visual inspection

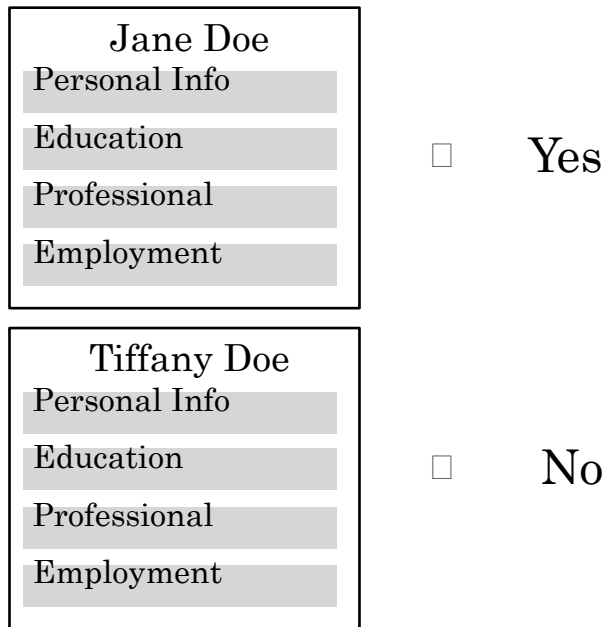
Recruiting

Data science



Optimize recruiting funnel

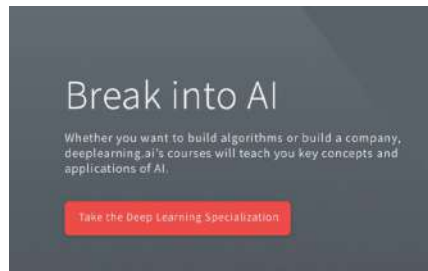
Machine learning



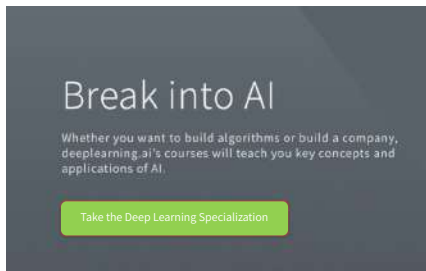
Automated resume screening

Marketing

Data science



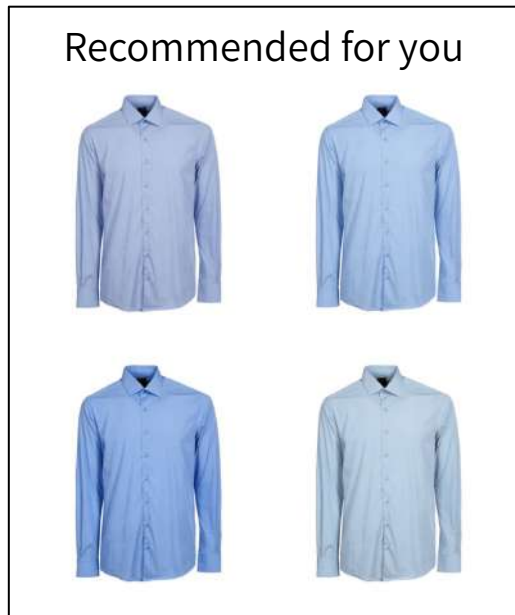
A



B

A/B testing

Machine learning



Customized product recommendation

Agriculture

Data science



Crop analytics

Machine learning



Precision weed killing



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Building AI Projects

How to choose an AI project I

AI knowledge and domain knowledge

What AI
can do

Things
valuable
for your
business

AI experts

Domain experts



Brainstorming framework

- Think about optimizing tasks rather than automating jobs. E.g., call center routing, radiologists.
- What are the main drivers of business value?
- What are the main points in your business?

You can make progress even without big data

- Having more data almost never hurts.
- Data makes some businesses (like web search) defensible.
- But with small datasets, you might still make progress.





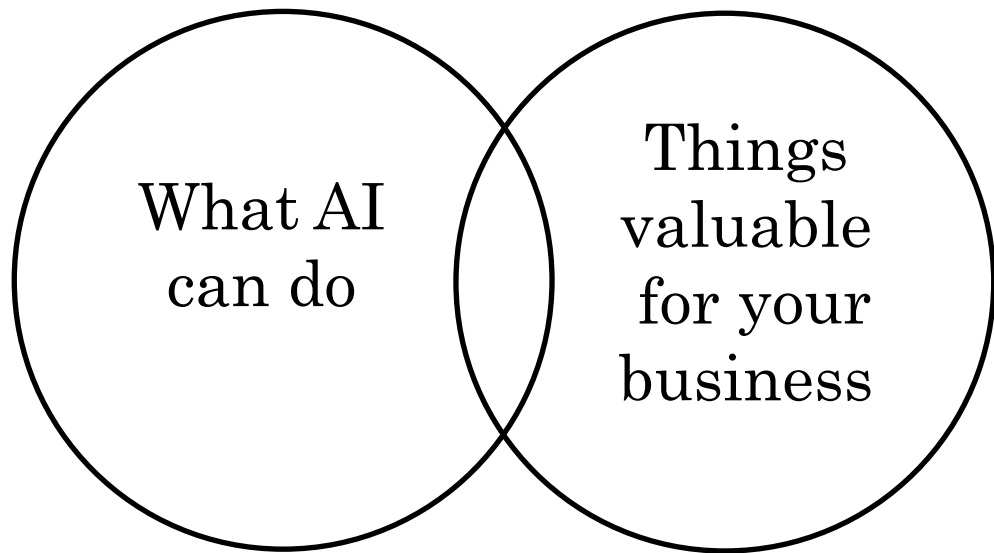
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Building AI Projects

How to choose an AI project II

Due diligence on project



Technical
diligence

Business
diligence

Due diligence on project

Technical diligence

- Can AI system meet desired performance
- How much data is needed
- Engineering timeline

Business diligence

- Lower costs current business
- Increase revenue current business
- Launch new product or business new business

Build vs. buy

- ML projects can be in-house or outsourced
- DS projects are more commonly in-house
- Some things will be industry standard – avoid building those.



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Building AI Projects

Working with an AI team

Specify your acceptance criteria



ok

Goal: detect defects
with 95% accuracy



ok

Provide AI team a dataset on
which to measure their
performance



defect

How AI teams think about data

Training set



ok



ok



ok

Test set



ok



ok



defect

Pitfall: Expecting 100% accuracy

Test set



ok



ok



ok



defect



ok

- Limitations of ML
- Insufficient data
- Mislabeled data
- Ambiguous label



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Building AI Projects

Technical tools for
AI teams (optional)

Open-source frameworks

Machine learning frameworks:

- TensorFlow
- PyTorch
- Keras
- MXNet
- CNTK
- Caffe
- PaddlePaddle
- Scikit-learn
- R
- Weka

Research publications

- Arxiv

Open source repositories:

- GitHub

CPU vs. GPU

CPU: Computer processor (Central Processing Unit)



GPU: Graphics Processing Unit



Cloud vs. On-premises

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Building AI in Your Company

Introduction

Building AI in your company

- Case studies of complex AI products
- Roles in an AI team
- AI Transformation Playbook
- Taking your first step



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Building AI in Your Company

Case study: Smart speaker

Smart speaker



Amazon
Echo / Alexa



Google
Home



Apple
Siri



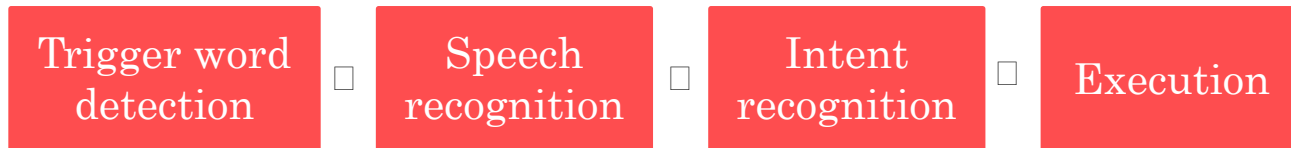
Baidu
DuerOS

“Hey device, tell me a joke”

“Hey device, tell me a joke”

Steps to process the command:

- | | | | |
|----|---------------------------------|------------------|---|
| 1. | Trigger word/wakeword detection | Audio | “Hey device”? (0/1) |
| 2. | Speech recognition | Audio | “tell me a joke” |
| 3. | Intent recognition | “tell me a joke” | joke?
time?
music?
call?
weather? |
| 4. | Execute joke | | |



“Hey device, set timer for 10 minutes”

Steps to process the command:

1. Trigger word/wakeword detection Audio “Hey device”? (0/1)
2. Speech recognition Audio “set timer for 10 minutes”
3. Intent recognition “set timer for 10 minutes” timer
4. a) Extract duration
 “Set timer for 10 minutes”
 “Let me know when 10 minutes is up”
 b) Start timer with set duration

Other functions

- Play music
- Volume up/down
- Make call
- Current time
- Units conversion
- Simple question
- ...

Key steps:

1. Trigger/wakeword detection
2. Speech recognition
3. Intent recognition
4. Specialized program to execute command



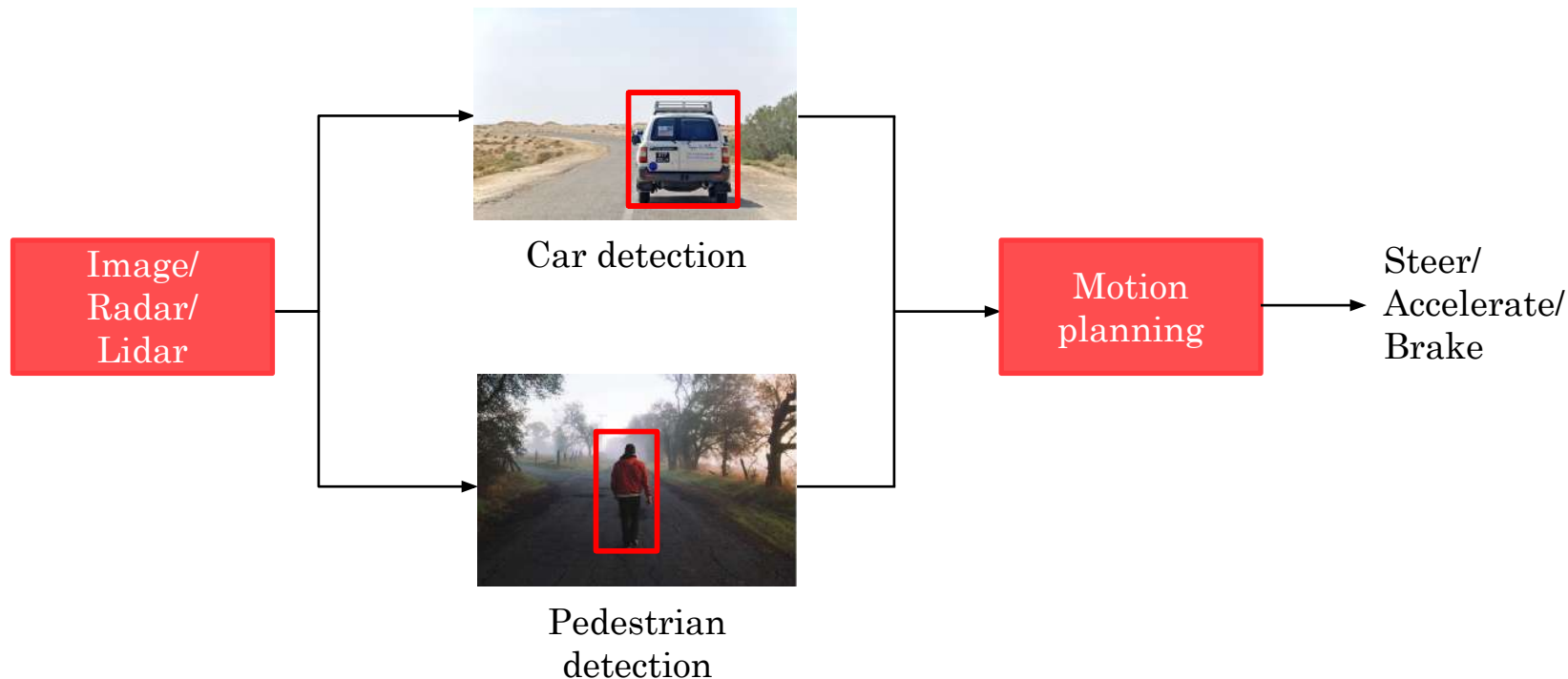
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Case study: Self-driving car

Steps for deciding how to drive



Key steps:

1. Car detection



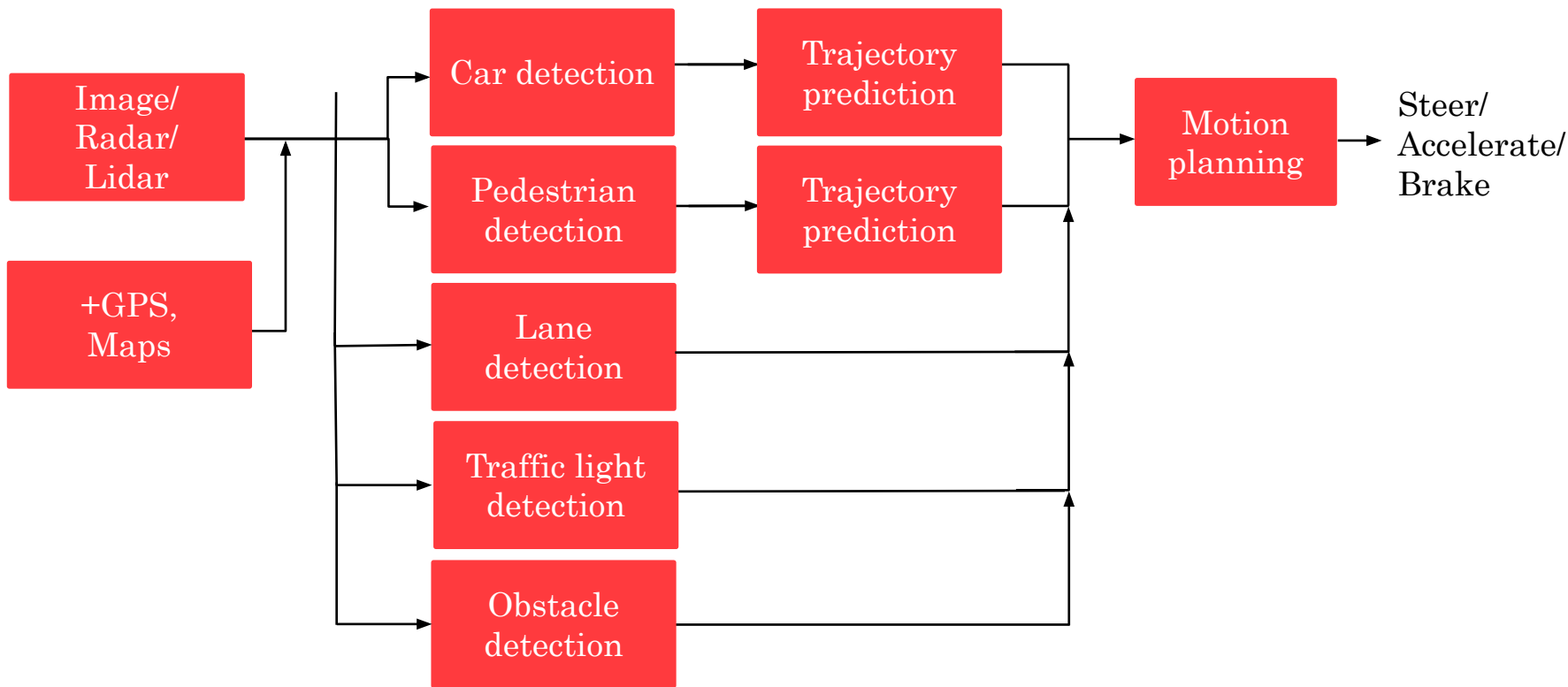
2. Pedestrian detection



3. Motion planning



Steps for deciding how to drive





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Building AI in Your Company

Example roles of an AI team

Example roles

- Software Engineer

- E.g., joke execution, ensure self-driving reliability, ...

- Machine Learning Engineer

Applied ML Scientist

- Machine Learning Researcher

- Extend state-of-the-art in ML

Example roles

- Data Scientist

- Examine data and provide insights
- Make presentation to team/executive

- Data Engineer

- Organize data
- Make sure data is saved in an easily accessible, secure and cost effective way

1 MB (megabyte)

1,000 MB = GB (gigabyte)

1,000,000 MB = TB (terabyte)

1,000,000,000 MB = PB (petabyte)

- AI Product Manager

- Help decide what to build; what's feasible and valuable

Getting started with a small team

- 1 Software Engineer , or
- 1 Machine Learning Engineer/Data Scientist , or
- Nobody but yourself



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Building AI in Your Company

AI Transformation Playbook
(Part I)

AI Transformation Playbook

1. Execute pilot projects to gain momentum
2. Build an in-house AI team
3. Provide broad AI training
4. Develop an AI strategy
5. Develop internal and external communications

1. Execute pilot projects to gain momentum

- More important for the initial project to succeed rather than be the most valuable
- Show traction within 6-12 months
- Can be in-house or outsourced

2. Build an in-house AI team

BU= Business Unit

AI function can be under CTO, CIO, CDO, etc. or a new CAIO

3. Provide broad AI training

Role	What they should learn
Executives and senior business leaders	• What AI can do for your enterprise • AI strategy • Resource allocation
Leaders of divisions working on AI projects	• Set project direction (technical and business diligence) • Resource allocation • Monitor progress
AI engineer trainees	• Build and ship AI software • Gather data • Execute on specific AI projects

The smart CLO knows they should *curate* rather than *create* content



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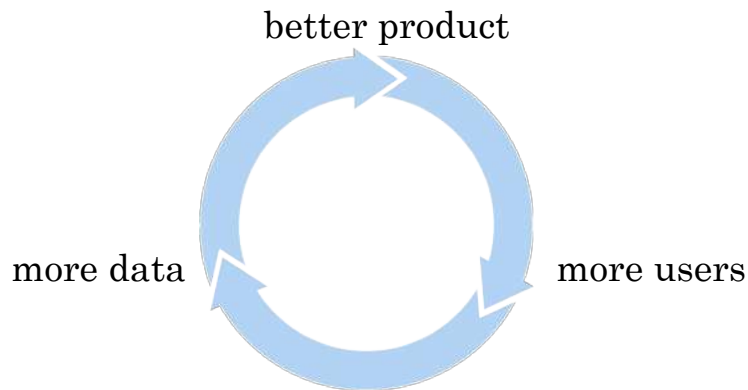
AI Transformation Playbook
(Part II)

AI Transformation Playbook

1. Execute pilot projects to gain momentum
2. Build an in-house AI team
3. Provide broad AI training
4. Develop an AI strategy
5. Develop internal and external communications

4. Develop an AI strategy

- Leverage AI to create an advantage specific to your industry sector
- Design strategy aligned with the “Virtuous Cycle of AI”



4. Develop an AI strategy

- Consider creating a data strategy
 - Strategic data acquisition
 - Unified data warehouse
 - Create network effects and platform advantages
 - In industries with “winner take all” dynamics,
- AI can be an accelerator

5. Develop internal and external communications

- Investor relations
- Government relations
- Customer/user education
- Talent/recruitment
- Internal communications

Detailed AI Transformation Playbook: <https://landing.ai/ai-transformation-playbook/>



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Building AI in Your Company

AI pitfalls to avoid

AI pitfalls to avoid

Don't:

- Expect AI to solve everything
- Hire 2-3 ML engineers and count solely on them to come up with use cases

Do:

- Be realistic about what AI can and cannot do given limitations of technology, data, and engineering resources
- Pair engineering talent with business talent and work cross-functionally to find feasible and valuable projects

AI pitfalls to avoid

Don't:

- Expect the AI project to work the first time
- Expect traditional planning processes to apply without changes
- Think you need superstar AI engineers before you can do anything

Do:

- Plan for AI development to be an iterative process, with multiple attempts needed to succeed
- Work with AI team to establish timeline estimates, milestones, KPIs, etc.
- Keep building the team, but get going with the team you have



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Building AI in Your Company

Taking your first step in AI

Some initial steps you can take

- Get friends to learn about AI
 - This course
 - Reading group
- Start brainstorming projects
 - No project is too small
- Hire a few ML/DS people to help
- Hire or appoint an AI leader (VP AI, CAIO, etc.)
- Discuss with CEO/Board possibilities of AI Transformation
 - Will your company be much more valuable and/or more effective if it were good at AI?



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Building AI in Your Company

Survey of major AI
application areas
(optional)

Computer Vision

- Image classification/Object recognition
 - Face recognition



cat

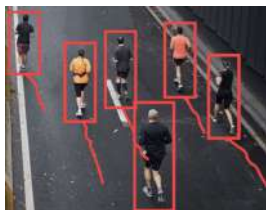
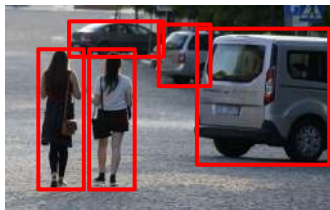
register



new



- Object detection
- Image segmentation
- Tracking



Natural Language Processing

- Text classification
 - Sentiment recognition
- Information retrieval
 - E.g., web search
- Name entity recognition
- Machine translation

Email

Product description

“The food was good”

“Service was horrible”

Spam/Non-Spam

Product category



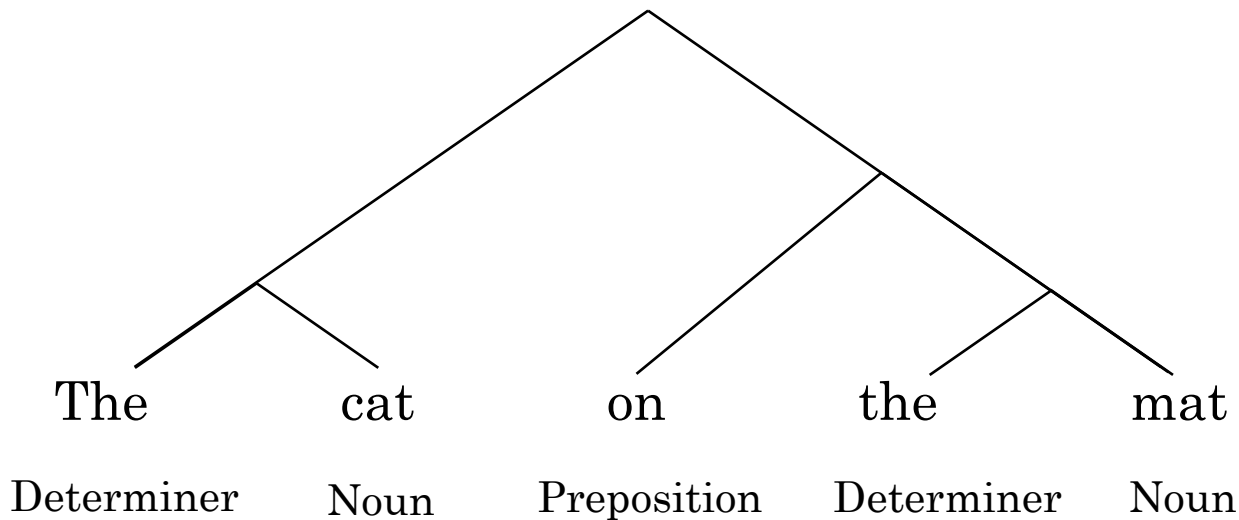
“Queen Elizabeth II knighted
Sir Paul McCartney for his
services to music at the
Buckingham Palace”

AIは、新たな電気だ

AI is the new electricity

Natural Language Processing

- Others: parsing, part-of-speech tagging



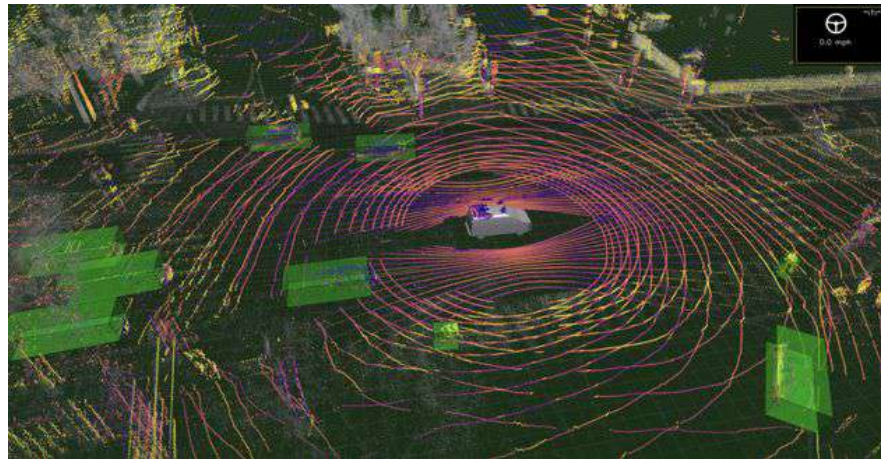
Speech



- Speech recognition (speech-to-text)
- Trigger word/wakeword detection
- Speaker ID
- Speech synthesis (text-to-speech, TTS)
The quick brown fox jumps over the lazy dog.

Robotics

- Perception: figuring out what's in the world around you
- Motion planning: finding a path for the robot to follow
- Control: sending commands to the motors to follow a path



General machine learning

- Unstructured data (images, audio, text)



image



audio

AIは、新たな電気だ

text

AI is the new electricity

- Structured data

House size (square feet)	# of bedrooms	Price (1000\$)
523	1	100
645	1	150
708	2	200

Clay batch #	Supplier	Mixing time (minutes)
001	ClayCo	35
034	GooClay	22
109	BrownStuff	28



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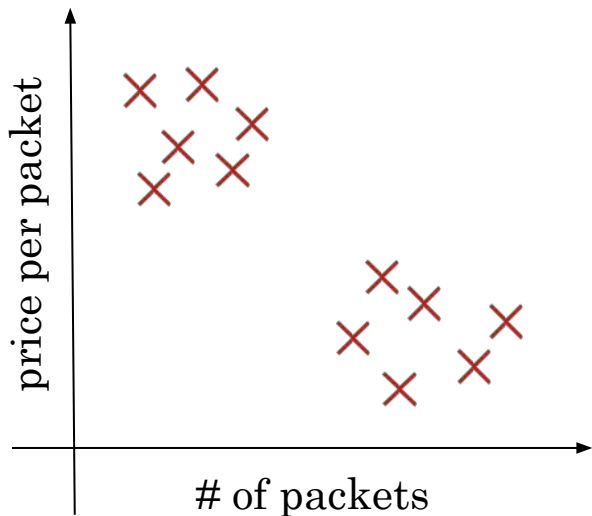
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Building AI in Your Company

Survey of major AI techniques
(optional)

Unsupervised learning

Clustering Potato chip sales



Given data (without any specific desired output labels), find something interesting about the data



Finding cats from
unlabeled YouTube videos

Transfer learning

Car detection



100,000 images

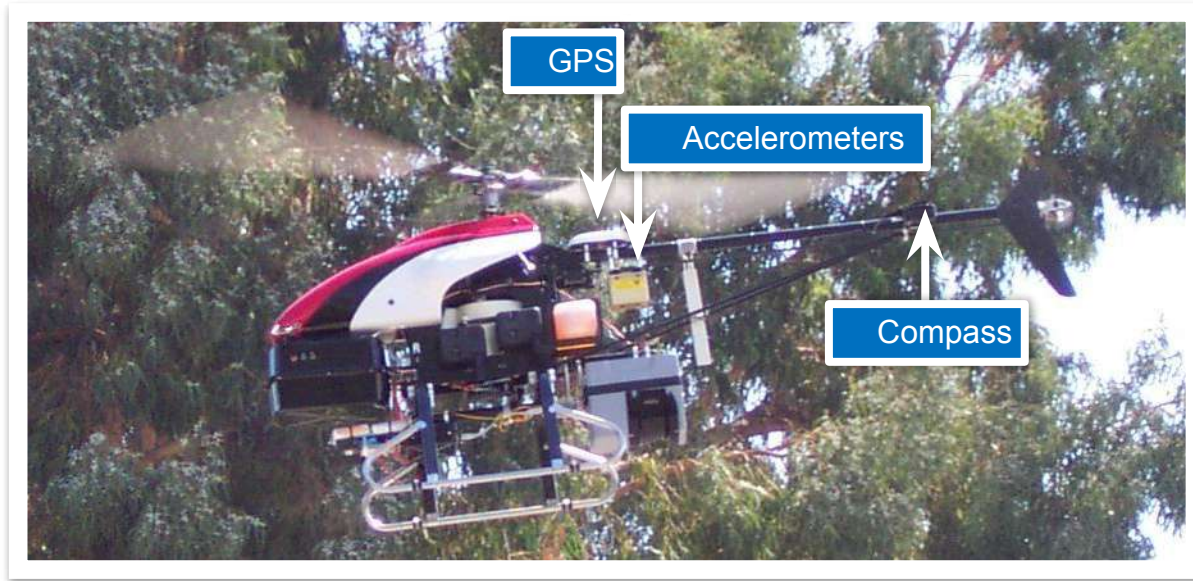
Golf cart detection



100 images

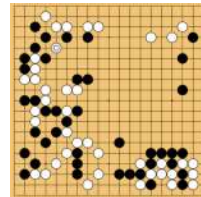
Learn from task A, and use knowledge to help on task B

Reinforcement learning



Use a “reward signal” to tell the AI when it is doing well or poorly. It automatically learns to maximize its rewards.

Reinforcement learning



Use a “reward signal” to tell the AI when it is doing well or poorly. It automatically learns to maximize its rewards.

GANs (Generative Adversarial Network)

Synthesize new images from scratch



[Source: Karras et al. (2018). Progressive Growing of GANs for Improved Quality, Stability, and Variation]

Knowledge Graph

leonardo da vinci

All Images Books News Videos More Settings Tools

ada lovelace

All Images Books Videos News More Settings Tools

About 9,290,000 results (0.35 seconds)

Ada Lovelace - Wikipedia
https://en.wikipedia.org/wiki/Ada_Lovelace
Augusta Ada King, Countess of Lovelace was an English mathematician and writer, chiefly known for her work on Charles Babbage's proposed mechanical ...
Resting place: Church of St. Mary Magdalene, Spouse(s): William King-Noel, 1st Earl of Lovelace
...
Known for: Mathematics, computing
Charles Babbage · Analytical Engine · William King-Noel, 1st Earl of · Lady Byron


More images

Ada Lovelace: Founder of Scientific Computing
<https://www.sdsc.edu/ScienceWomen/lovelace.html>
ADA BYRON, COUNTESS OF LOVELACE ... Ada Byron was the daughter of a brief marriage between the Romantic poet Lord Byron and Anne Isabelle ...

People also ask

What is Ada Lovelace famous for?

What did Ada Lovelace invent and what impact it had?

When did Ada Lovelace invent the computer?

What is Ada Lovelace Day?

Feedback

Ada Lovelace | Biography & Facts | Britannica.com
<https://www.britannica.com/biography/Ada-Lovelace>
Jan 3, 2019 - Ada Lovelace, in full Ada King, countess of Lovelace, original name Augusta Ada Byron, Lady Byron, born December 10, 1815, ...

Ada Lovelace
Mathematician

Augusta Ada King, Countess of Lovelace was an English mathematician and writer, chiefly known for her work on Charles Babbage's proposed mechanical general-purpose computer, the Analytical Engine. Wikipedia

Born: December 10, 1815, London, United Kingdom
Died: November 27, 1852, Marylebone, United Kingdom
Spouse: William King-Noel, 1st Earl of Lovelace (m. 1835–1852)
Children: Anne Blunt, 15th Baroness Wentworth, MORE
Parents: Lord Byron, Lady Byron
Known for: Mathematics, Computing

People also search for

View 15+ more

Ada Lovelace	
Born	Dec 10, 1815
Died	Nov 27, 1852
Bio	English mathematician and writer...

Northern Rooster Hotel	
Address	45 Rooster St, LA
Phone	(650) 555-3992
Wifi	yes
Pool	no

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AI and Society

Introduction

AI and society

- AI and hype
- Limitations of AI

- Bias

AI and Ethics

- Adversarial attacks

- AI, developing economies, and jobs
- Conclusion



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AI and Society

A realistic view of AI

Goldilocks rule for AI

- Too optimistic: Sentient / super-intelligent AI killer robots coming soon
- Too pessimistic: AI cannot do everything, so an AI winter is coming
- Just right: AI can't do everything, but will transform industries

Limitations of AI

- Performance limitations
- Explainability is hard (but sometimes doable)



Right-sided
Pneumothorax
(collapsed lung)



[Rajpurkar et al. (2018). CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning.]

[Wang et al. (2017). ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. IEEE CVPR]

[Images source: NIH Clinical Center Image dataset: <https://nihcc.app.box.com/v/ChestXray-NIHCC>]

Limitations of AI

- Biased AI through biased data
- Adversarial attacks on AI



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AI and Society

Discrimination / Bias

AI learning unhealthy stereotypes

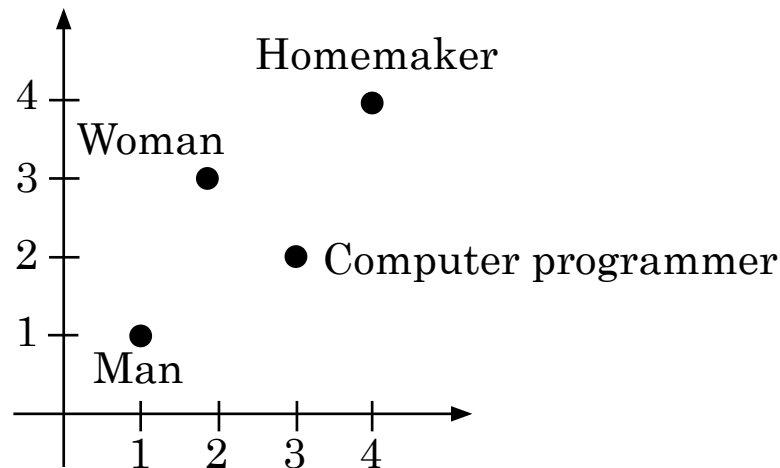
- **Man** : **Woman** as **Father** : **Mother**
- **Man** : **Woman** as **King** : **Queen**
- Man : Computer programmer as Woman : Homemaker

Man: (1,1)

Computer programmer: (3,2)

Woman: (2,3)

Homemaker: (4,4)



Bolukbasi et al. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings.

Why bias matters

- Hiring tool that discriminated against women
- Facial recognition matching dark skinned individuals to criminal mugshots
- Bank loan approvals
- Toxic effect of reinforcing unhealthy stereotypes

Combating bias

- Technical solutions:
 - E.g., “zero out” the bias in words
 - Use less biased and/or more inclusive data
- Transparency and/or auditing processes
- Diverse workforce
 - Creates less biased applications



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Adversarial attacks
on AI

Adversarial attacks on AI



Hummingbird

Minor perturbation →



Hammer



Hare

Minor perturbation →



Desk

Physical attacks



“Milla Jovovich”

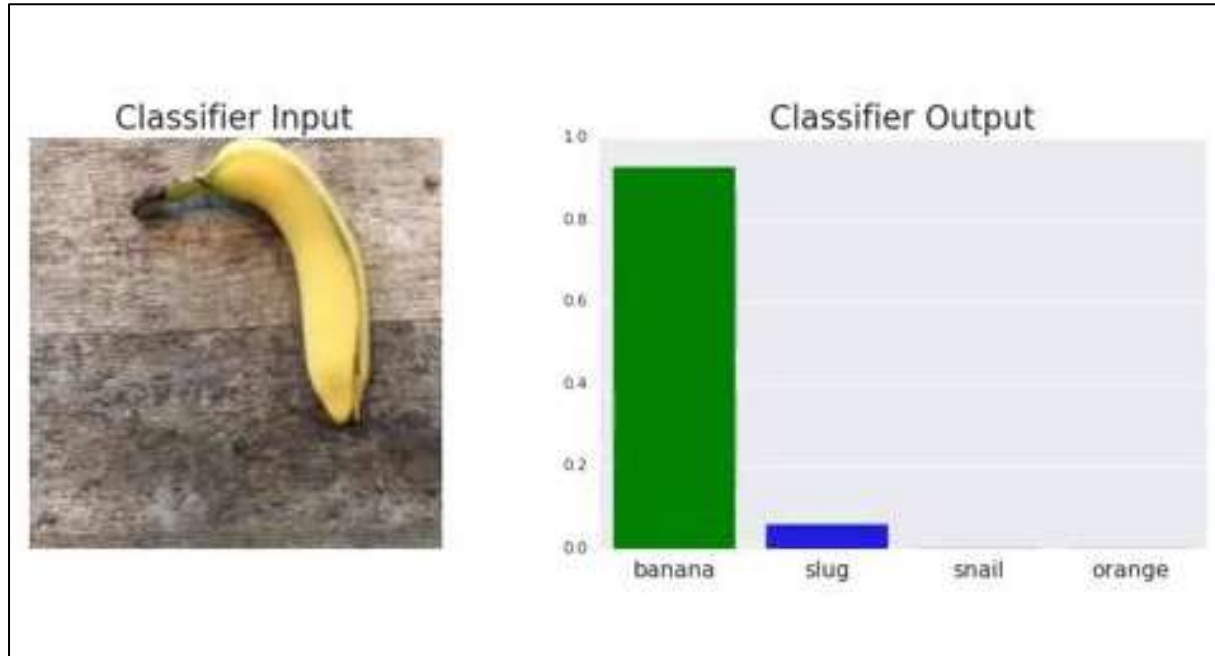


Fails to see stop sign



Banana

Physical attacks



[Sharif et al. (2016). Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition.]

[Eykholt et al. (2018). Physical Adversarial Examples for Object Detectors.]

[Brown et al. (2018). Adversarial Patch.]

Adversarial defenses

- Defenses do exist, but incur some cost
- Similar to spam vs. anti-spam, we may be in an arms race for some applications



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AI and Society

Adverse uses of AI

Adverse uses of AI

- DeepFakes
 - Synthesize video of people doing things they never did
- Undermining of democracy and privacy
 - Oppressive monitoring of individuals
- Generating fake comments
- Spam vs. anti-spam and fraud vs. anti-fraud



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AI and Society

AI and developing economies

Developing economies

“leapfrog”

- Mobile phones
- Mobile payments
- Online education

How developing economies can build AI

- US and China are leading, but all AI communities are still immature
- Focus on AI to strengthen a country's vertical industries
- Public-private partnerships to accelerate development



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AI and Society

AI and jobs

AI's impact on jobs worldwide

Jobs replaced
by 2030

400-800_{mil}

Jobs created
by 2030

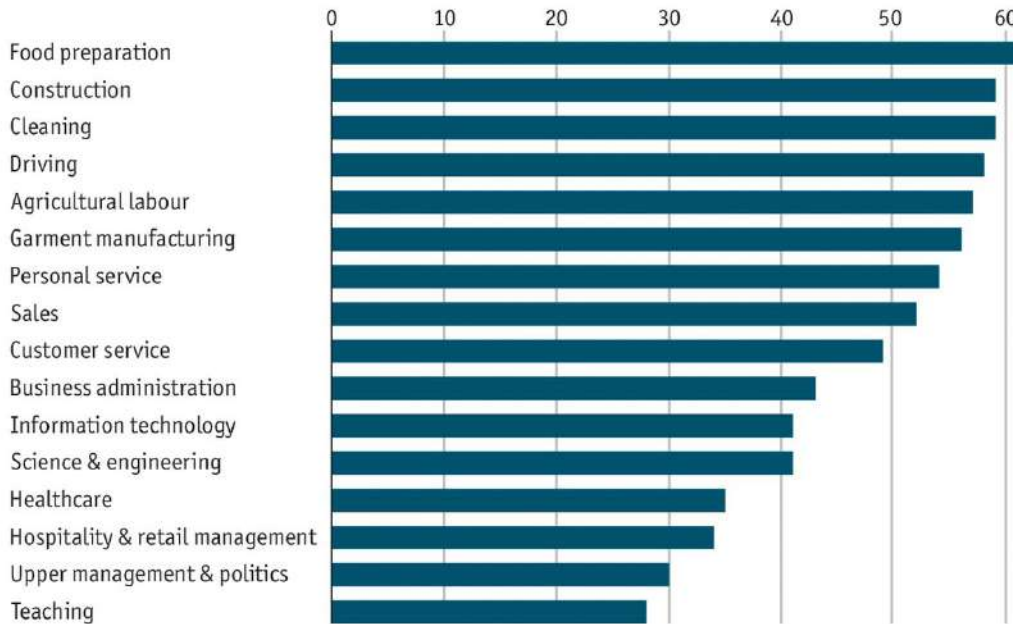
555-890_{mil}

[Source: McKinsey Global Institute.]

AI's impact on jobs worldwide

Automated for the people

Automation risk by job type, %



Source: OECD

Economist.com

[Image credit: Economist.com]

[Nedelkoska, L. and G. Quintini. (2018). Automation, skills use and training. *OECD Social, Employment and Migration Working Papers*, No. 202.]

Some solutions

- Conditional basic income: provide a safety net but incentivize learning
- Lifelong learning
- Political solutions



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AI for Everyone

Conclusion

What you've learned

- What is AI?
- Building AI projects
- Building AI in your company
- AI and society

Keep learning!

- Online courses, books, blogs, ...
- deeplearning.ai mailing list

Thank you!